

4.4 Electric and Magnetic Fields

This topic covers Coulomb's law, capacitors, magnetic flux density and the laws of electromagnetic induction. This topic may be studied using applications that relate to, for example, communications and display techniques.

Students will be assessed on their ability to:	Suggested experiments
83 explain what is meant by an electric field and recognise and use the expression electric field strength $E = F/Q$	
84 draw and interpret diagrams using lines of force to describe radial and uniform electric fields qualitatively	Demonstration of electric lines of force between electrodes
85 use the expression $F = kQ_1Q_2/r^2$, where $k = 1/4\pi\epsilon_0$ and derive and use the expression $E = kQ/r^2$ for the electric field due to a point charge	Use electronic balance to measure the force between two charges
86 investigate and recall that applying a potential difference to two parallel plates produces a uniform electric field in the central region between them, and recognise and use the expression $E = V/d$	
87 investigate and use the expression $C = Q/V$	Use a Coulometer to measure charge stored
88 recognise and use the expression $W = \frac{1}{2} QV$ for the energy stored by a capacitor, derive the expression from the area under a graph of potential difference against charge stored, and derive and use related expressions, for example, $W = \frac{1}{2} CV^2$	Investigate energy stored by discharging through series/parallel combination of light bulbs
89 investigate and recall that the growth and decay curves for resistor–capacitor circuits are exponential, and know the significance of the time constant RC	

Students will be assessed on their ability to:	Suggested experiments
90 recognise and use the expression $Q = Q_0 e^{-t/RC}$ and derive and use related expressions, for exponential discharge in RC circuits, for example, $I = I_0 e^{-t/RC}$	Use of data logger to obtain I-t graph
91 explore and use the terms magnetic flux density B , flux Φ and flux linkage $N\Phi$	
92 investigate, recognise and use the expression $F = BIl \sin \theta$ and apply Fleming's left hand rule to currents	Electronic balance to measure effect of I and l on force
93 recognise and use the expression $F = Bqv \sin \theta$ and apply Fleming's left hand rule to charges	Deflect electron beams with a magnetic field
94 investigate and explain qualitatively the factors affecting the emf induced in a coil when there is relative motion between the coil and a permanent magnet and when there is a change of current in a primary coil linked with it	Use a data logger to plot V against t as a magnet falls through a coil of wire
95 investigate, recognise and use the expression $\varepsilon = -d(N\Phi)/dt$ and explain how it is a consequence of Faraday's and Lenz's laws	